# Madhab Chandra Das

**Digital Forensics and Cybersecurity Resume**

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## Professional Summary

Digital and cyber forensics PhD student with a master's degree in Cybersecurity and hands-on training in Windows artifact analysis, forensic imaging, deleted file recovery, packet analysis, and e-discovery. Uses Python, Autopsy, FTK Imager, Wireshark, TShark, ALEAPP, and Volatility in coursework and research. Builds structured datasets and evaluates machine learning and LLM outputs against evidence. Brings IT support experience and banking experience in due diligence, customer problem solving, staff training, and leadership of teams of 11–15 members.

## Technical Skills

Forensic tools: Autopsy, FTK Imager, HxD, John the Ripper, Wireshark, TShark, ALEAPP, Volatility 2 and 3.

Programming and systems: Python, PowerShell, SQL, Linux, VirtualBox, Kali Linux.

AI and analysis: Random Forest, XGBoost, LightGBM, CatBoost, Logistic Regression, SHAP, permutation importance, LLM APIs.

## Practical Forensic Training

### Windows and Operating System Forensics

Forensic imaging deleted file recovery, metadata and hex examination, password recovery, Registry and event log analysis, USB and browser artifacts, email headers, and HAR analysis.

### Network Forensics

Wireshark and TShark analysis of TCP connections, handshakes, visible payloads, login-related information, and suspicious traffic.

### Mobile and Memory Forensics

Android artifact examination with ALEAPP and memory analysis using Linux scripting.

### E Discovery

Case exercises, record preservation, TAR, METS, and procedural rules. Tools: PST Analyzer, MBOX Viewer, GoldFynch.

## Academic Experience

**Graduate Research Assistant and Teaching Assistant**

Sam Houston State University, Department of Computer Science | August 2025–Present

- Conduct forensic research using Python, artifact examination, structured datasets, and evaluation of AI and machine learning outputs.
- Conduct lab classes, proctor examinations, grade exam responses, and provide tutoring outside scheduled class hours.

## Education

**PhD in Digital and Cyber Forensics | Sam Houston State University**

August 2025–Present | Expected July 2029 | Current CGPA: 4.00

**Master's in Cybersecurity | Washington University of Science and Technology**

October 2023–June 2025 | CGPA: 3.92/4.00

MBA | Jahangirnagar University | August 2020–July 2022

BSc in Electrical and Electronic Engineering | UITS | August 2008–January 2013

# Research Projects

## Android Forensic Reporting

Examined public Android 13 data and a physical LG device using Python and ALEAPP. Generated reports through three LLM APIs and evaluated 612 claims across six reports against verified findings.

## Network Traffic Analysis

Prepared two encrypted traffic datasets and evaluated four ML models. Analyzed 41,238 external false positives (4.35% FPR) using SHAP and permutation importance.

## Windows Memory Forensics

Built 1,899 process records with 18 features from 45 memory dumps using Volatility 2.6. Evaluated three ensembles with dump-grouped validation against rule-derived labels.

## Quantum Security and Forensics

Generated two physics-inspired synthetic QKD datasets with 5 million records each. Implemented classifiers and examined reliability under changed simulation conditions.

# Additional Professional Experience

**The City Bank Ltd., Bangladesh | March 2017–July 2022**

Progressed through SPMO and GL, Senior Officer and Team Leader, Assistant Manager, and Associate Manager roles.

Experience in banking due diligence, customer relationships, staff training, and leadership of 11–15 team members.

Mutual Trust Bank Ltd. | Senior Sales Officer, contractual | September 2016–February 2017

BRAC Bank Ltd. | Trainee Officer, contractual | June 2013–June 2016

BRAC Bank Ltd. | Technology Intern | June–September 2011

# Relevant Graduate Coursework

Operating System Forensics, Network Forensic Analysis, Live System & Memory Forensics, E-Discovery, Mobile Device Forensics, and Cryptography & Steganography.

# Training and Professional Development

## Cybersecurity and Cloud

Cybersecurity and Risk Management Specialist — PIIT | 2024

Getting Started with AWS Cloud Essentials — AWS through W3Schools | 2024

Learning Cryptography and Network Security — LinkedIn Learning | 2026

Wireshark: Functionality — LinkedIn Learning | 2026

## Artificial Intelligence and Quantum Computing

Introduction to Quantum Computing — LinkedIn Learning | 2026

## Practical Participation

Capture the Flag Security Event — Cisco–SHSU | February 2026.

# Honors and Leadership

- COSET PhD Digital & Cyber Forensics Scholarship, SHSU, 2025–2026. Renewed for 2026–2027.
- WUST Presidential Honor: Summer 2024, Fall 2024, and Spring 2025.
- President, Bangladesh Student Association, SHSU | August 2025–Present. Organize events and support incoming students.